

- 4 ai Waves maintain a constant phase difference with respect to each other (same phase difference at all times). **B1**
- aii Waves are of single wavelength / frequency. **B1**
- bi Waves from two slits can overlap and interfere. The resultant displacement at any point at any instant is the vector sum of the individual displacements caused by each wave. **B1**
- At positions where the two waves arrive in phase, constructive interferences occur, resulting in high-intensity fringes / maxima. **B1**
- At positions where the two waves arrive out of phase, destructive interferences occur, resulting in zero-intensity fringes / minima. **B1**
- bii $12 \Delta y = 26 \Rightarrow \Delta y = 2.167 \text{ mm}$ **C1**
- $$\Delta y = \frac{\lambda L}{d} \Rightarrow 2.167 \times 10^{-3} = \frac{\lambda(1.20)}{0.38 \times 10^{-3}}$$
- C1**
- $\lambda = 686 \text{ nm}$ **A1**
- biii Calculate average value of fringe separation to minimize random error or Reduces fractional / percentage uncertainty in the measurement of fringe width **B1**
- c Light reaches the eye through two paths – directly and after reflection off the mirror. **B1**
- They act like two sources of light which can interfere with each other to produce interference pattern. **B1**
- 5 ai An electric field is a region of space where a stationary charge experiences a force. **B1**